

Forecasting the stock market with LSTMs

Final degree project

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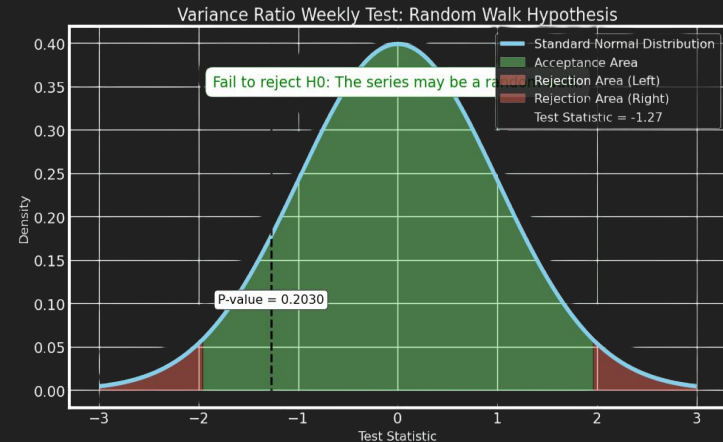
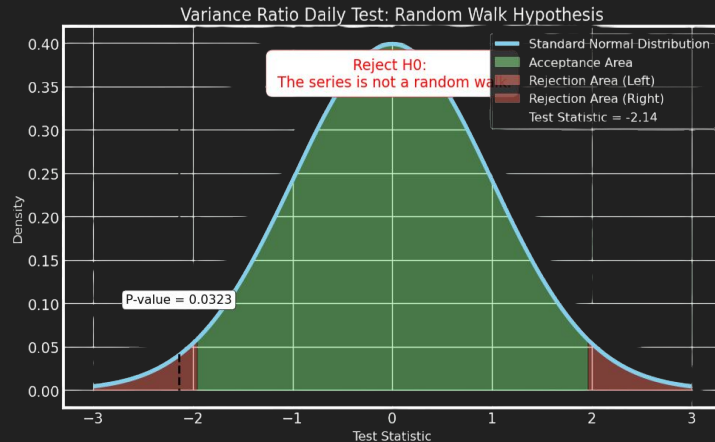
Pamplona, 24th January 2025



Random Walk: can markets be predicted?

Expert 1: “Humans are moronic pattern matching chimps and look for trends whether they are there or not.” (Efficient Market Hypothesis)

Expert 2: “God does not play dice with the universe. The stock market is no exception”. (Technical Analyst)



RNN for time series forecasting

RNN is designed to interpret temporal or sequential information

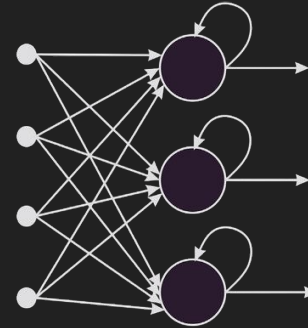
Recurrent connections $\Rightarrow$ Memory

Challenges and solutions

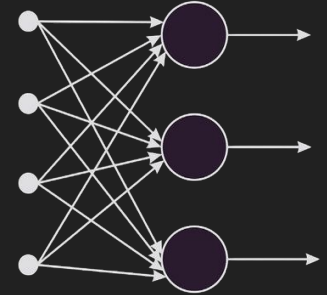
- Non-convergence (exploding gradients) $\Rightarrow$ Clipping gradients
- Short memory (vanishing gradients) $\Rightarrow$ Activation functions

Improvements:

- Gating mechanisms $\Rightarrow$ Improved gradient flow, better memory retention



(a) Recurrent Neural Network



(b) Feed-Forward Neural Network

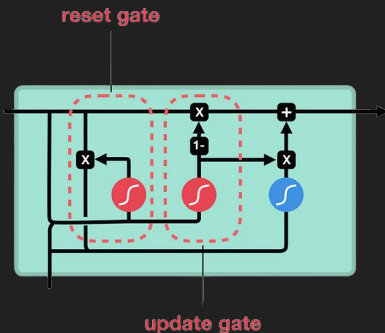
[ResearchGate]

RNN variants: GRU and LSTM networks

Gated Recurrent Unit (GRU):

- Hidden state
- Reset gate: what to forget
- Update gate: what to retain

More efficient, less prone to overfitting

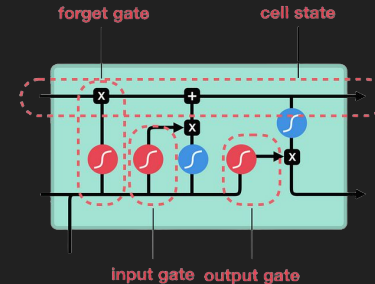


[Michael PHI. Towards Data Science]

Long Short Term Memory (LSTM):

- Cell state + Hidden state
- Forget gate: what to forget
- Input gate: what to retain
- Output gate: filter the output

Better long-term memory



[Michael PHI. Towards Data Science]

Baseline models

Naive model:

- Drift model. Project historical trend into the future.

Statistical model:

- SARIMA. AR and MA components. Handles seasonality and non-stationarity.

Machine Learning models:

- XGBoost. Decision-tree based ensemble, sequential error correction (boosting).
- ARNN: simplification of RNN. No hidden state, lagged values as features.

Standard & Poor 500 Index

Indices over stocks

- Reduced noise $\Rightarrow$ More reliable patterns

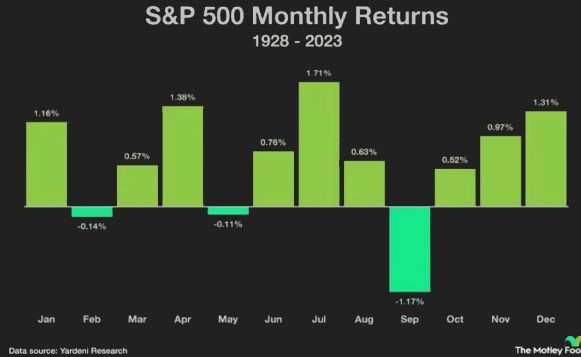
S&P 500 (500 US companies):

- Old data records
- Very high participation
- Diversification
- Related to US economy
- Related to US currency

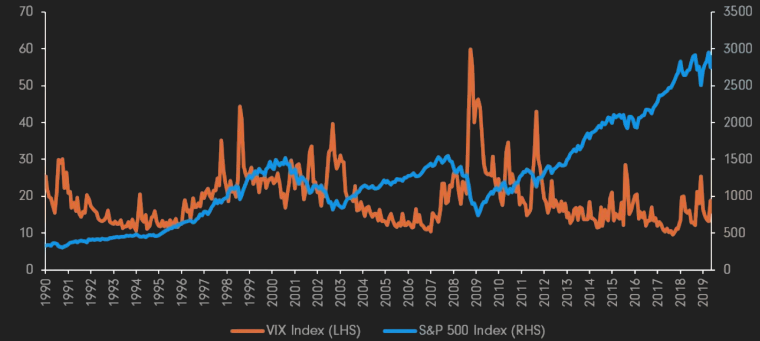


SPX composition at 23/01/2025. [Finviz]

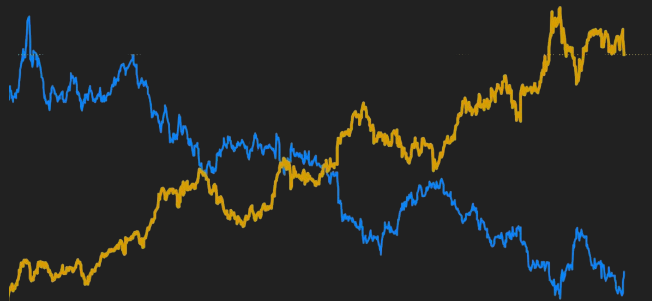
Data preparation (I): Exogenous variables



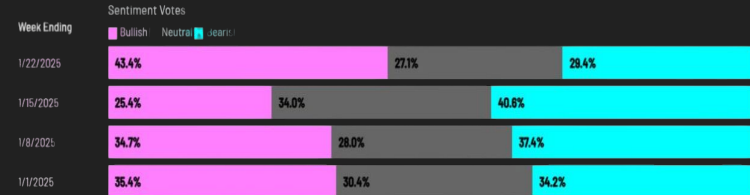
Historical monthly returns [The Motley Fool]



Market Volatility Index [Fidelity Hong Kong]



DXY [Rosewood Research]

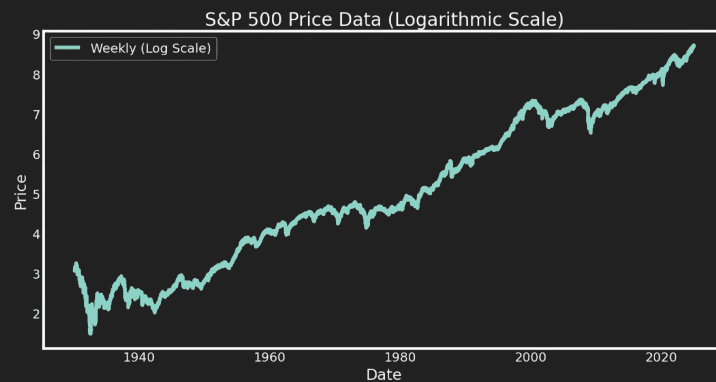
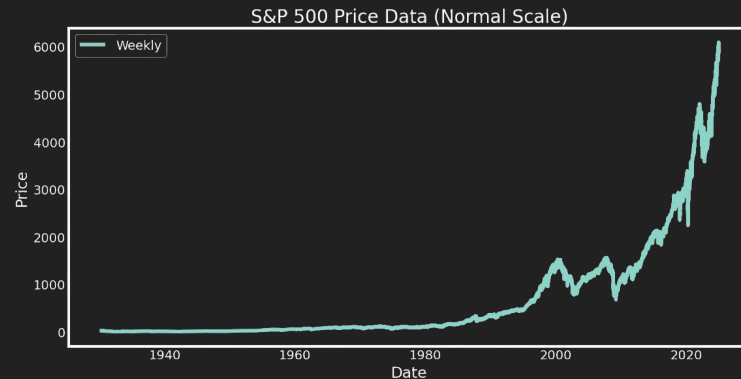
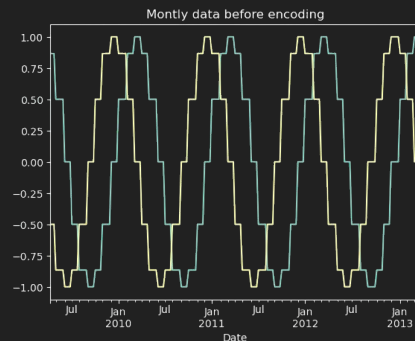
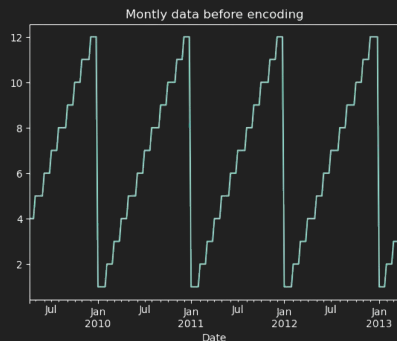


Investor sentiment [AAII Index]

Data preparation (II): Static preprocessing

General treatment:

- Logarithmic scale
- Investor sentiment scaling (/100)
- Month cyclical encoding (sin/cos)



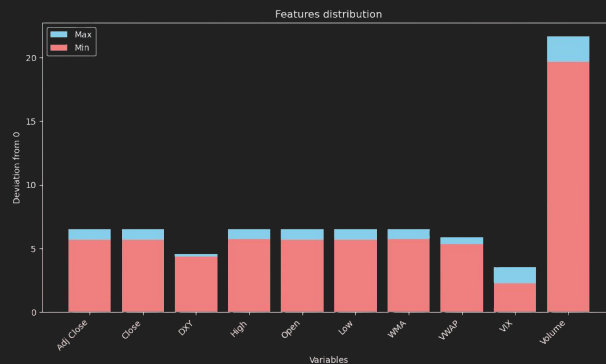
Data preparation (III): Dynamic preprocessing

Issue with static ML treatment:

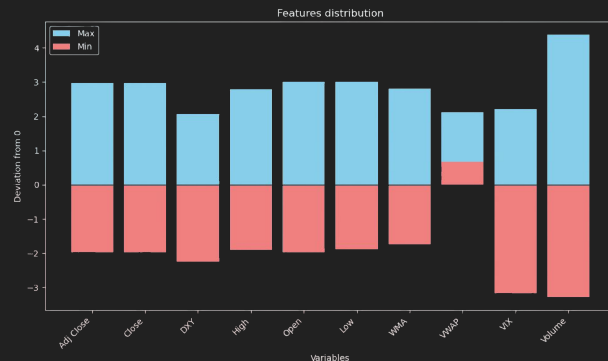
- Frequent new all-time highs $\Rightarrow$ Extrapolation problem

Solution: Sliding window

1. Removing trend: detrender
2. Similar ranges and distributions of features: scalers, transformers

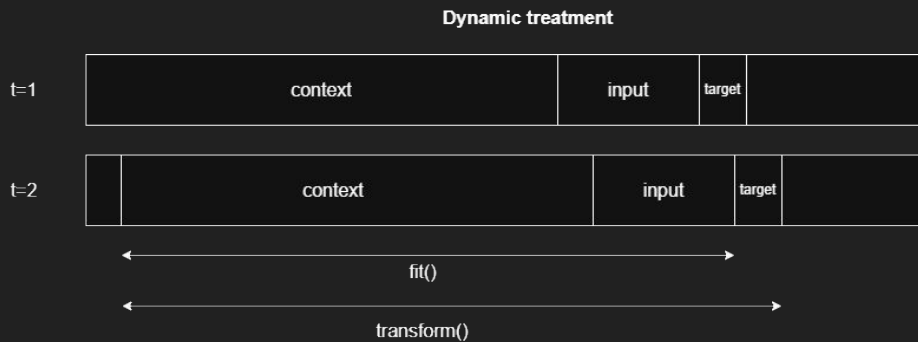


Ranges before dynamic treatment



Ranges after dynamic treatment

Data preparation (IV): Data split

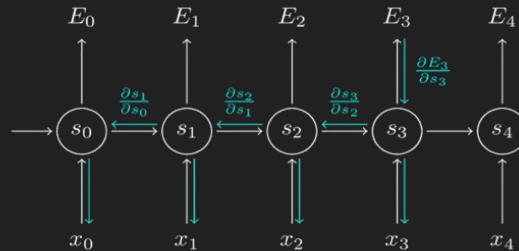


Backward Propagation Through Time (BPTT)

BPTT is designed to handle sequential data

It unfolds the RNN across time steps, keeping temporal dependencies.

Weights are adjusted based on how much each step contributed to the final loss



Backpropagation Through Time

[Medium]

Considerations when training and validating

Before each epoch:

- Reset hidden state $\Rightarrow$ Stateful model

Before each backpropagation:

- Detaching hidden state $\Rightarrow$ Not calculating gradients
- Gradients set to 0 $\Rightarrow$ Avoid accumulation from previous iterations

Walk-forward validation

- Dependent data points $\Rightarrow$ Predict n steps, update context with new information, repeat

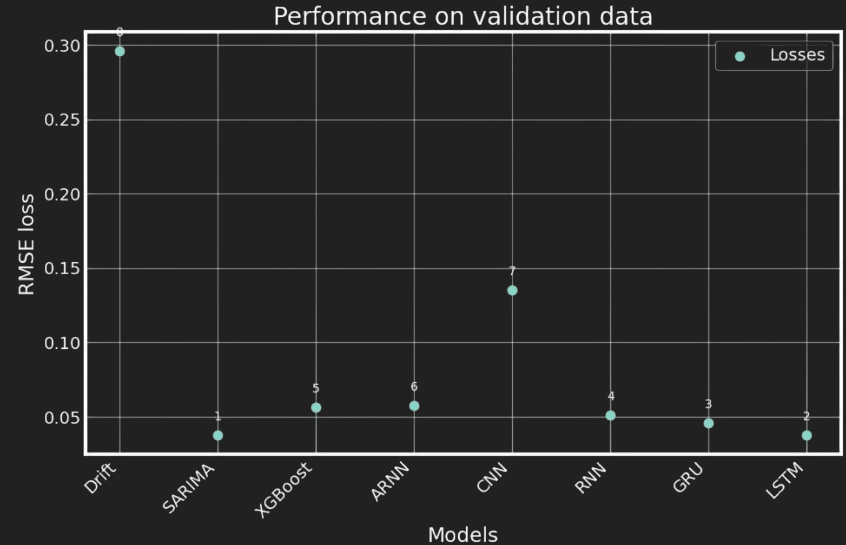
Optimization step 1: Validating models

Metric: RMSE Loss

- Interpretability

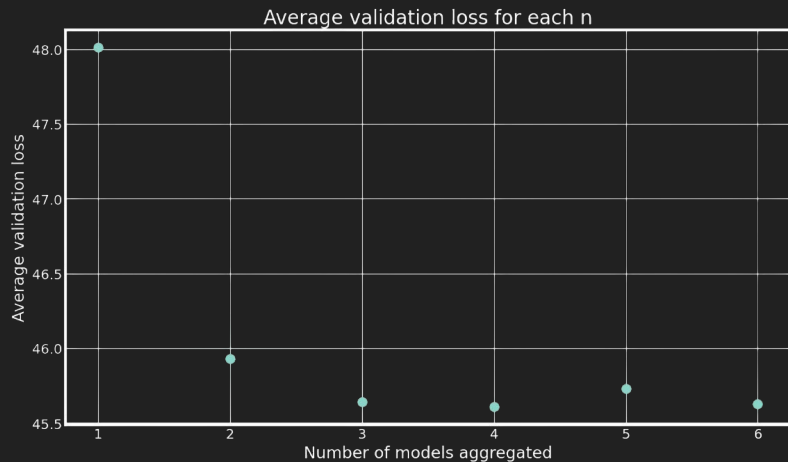
Model	Learning rate	Dropout	N layers	Hidden units
RNN	0.0002	0.645	1	216
GRU	0.0003	0.682	1	142
LSTM	0.0009	0.529	1	264

Best model-related parameters

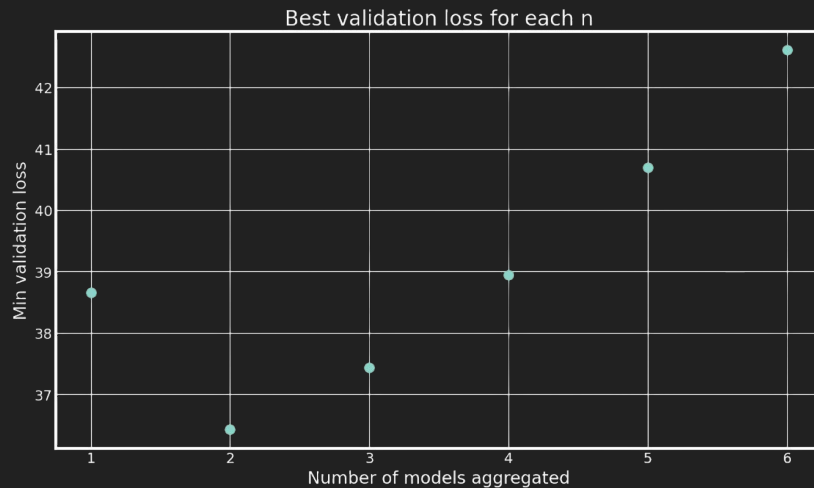


Performance on validation data

Optimization step 2: Aggregating predictions



Average validation loss for each n



Best validation loss for each n

Agent decision-making framework

Predicted change considered significant if greater than confidence threshold:

- Fixed differential (e.g. 2%)
- Dynamic differential (e.g. $1.5 \times \text{ATR}$)

Supporting indicators to identify:

- RSI (Relative Strength Index): overbought or oversold asset
- MACD (Moving Average Convergence/Divergence): regime shift based on moving averages
- OBV (On Balance Volume): increase in participation in the direction of the trend

Optimization step 3: Tuning the strategies

Metric: Sharpe Ratio

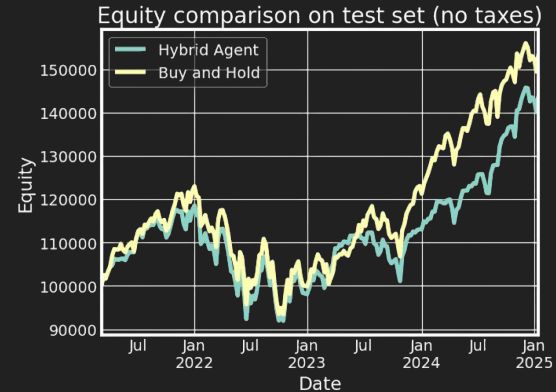
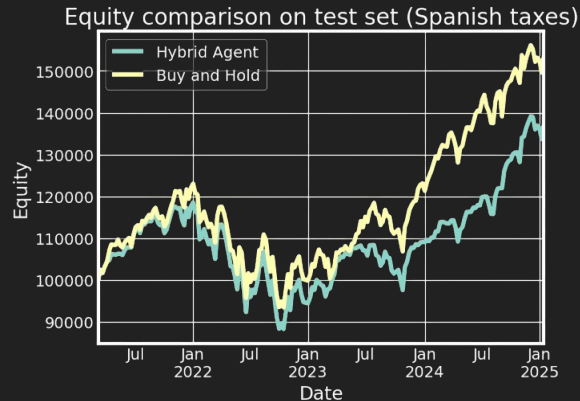
- Risk-free rate of return
- Industry standard

Strategy	Agg func	Close date	Threshold	Short ord
Predictions	Max	Variable	1,18xATR	No
Predictions + Indicators	Harmonic mean	Variable	0,62xATR	No

Strategy	Profit	PF	MDD	WR	Sharpe ratio
Predictions	32,81%	1,57	27,06%	70,58%	0,179
Predictions + Indicators	48,22%	4,86	27,78%	72,09%	0,191
Reference	60,25%	>1,75	<20%	>60%	>1

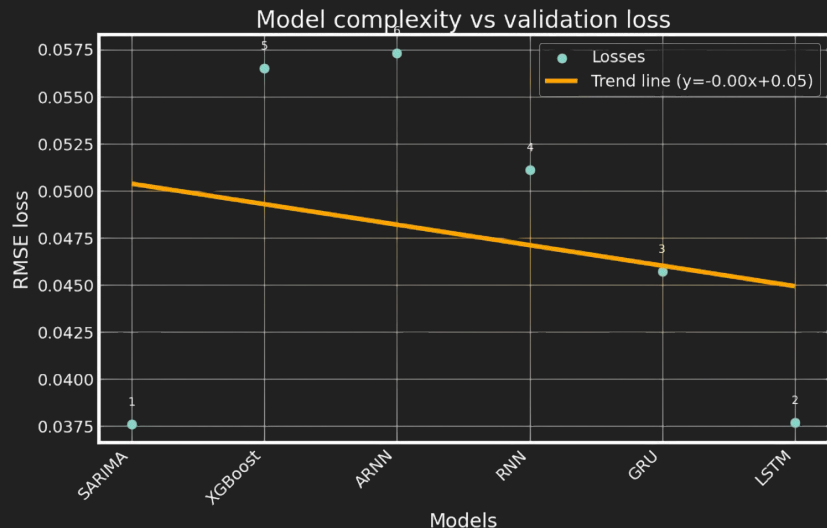
Final results: test metrics

Strategy	Profit	PF	MDD	WR	Sharpe ratio
Agent	37,36%	3,41	25,5%	80%	0,148
Tax free agent	41,56%	3,82	18,27%	82,35%	0,167
Reference	60,25%	>1,75	<20%	>60%	>1



Learnings and conclusions

Deep Learning for time series forecasting



Challenges with forecasting markets

- Too many variables
- No significant dependence

Difficulties of heuristic strategies

- Real decision making is hard to recreate

Complex, superrapid integrations can occur beneath conscious awareness and present themselves as “intuitions.”

Future lines

Pattern recognition (unsupervised learning):

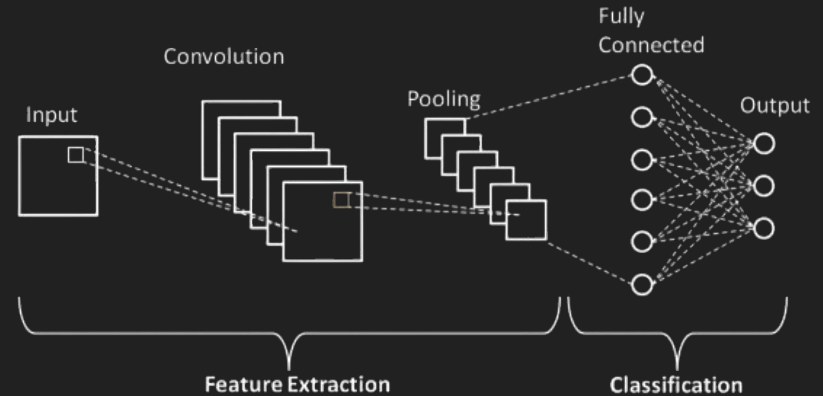
- CNN (Convolutional Neural Network)

Transformer-based models:

- TiDE (Time-series Dense Encoder)

Cross-disciplinary symbiosis:

- Fundamental analysis
- Integration of socioeconomic theory



CNN structure. [ResearchGate]